

PE-EC703A — Embedded System

Semester VII • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Embedded Hardware — Processor & Memory

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Contents

1	8051 Microcontroller — Architecture Overview	2
1.1	Key Features of 8051	2
1.2	Real-World Interfacing of 8051	2
2	ARM Processor Architecture	3
2.1	ARM vs. 8051 — Key Differences	3
2.2	ARM Processor Features	3
3	Processor and Memory Organization	4
3.1	Memory Types in Embedded Systems	4
3.2	Harvard vs. Von Neumann Architecture	4
3.3	Memory Map of a Typical MCU	4
4	Parallelism in Instruction Level	5
4.1	Pipelining	5
4.2	Pipeline Hazards	5
5	Processor and Memory Selection	6
5.1	Memory Selection Criteria	6

1. 8051 Microcontroller — Architecture Overview

The Intel 8051 is an 8-bit microcontroller introduced in 1980. It is one of the most widely taught architectures in embedded systems education due to its simplicity and self-contained design.

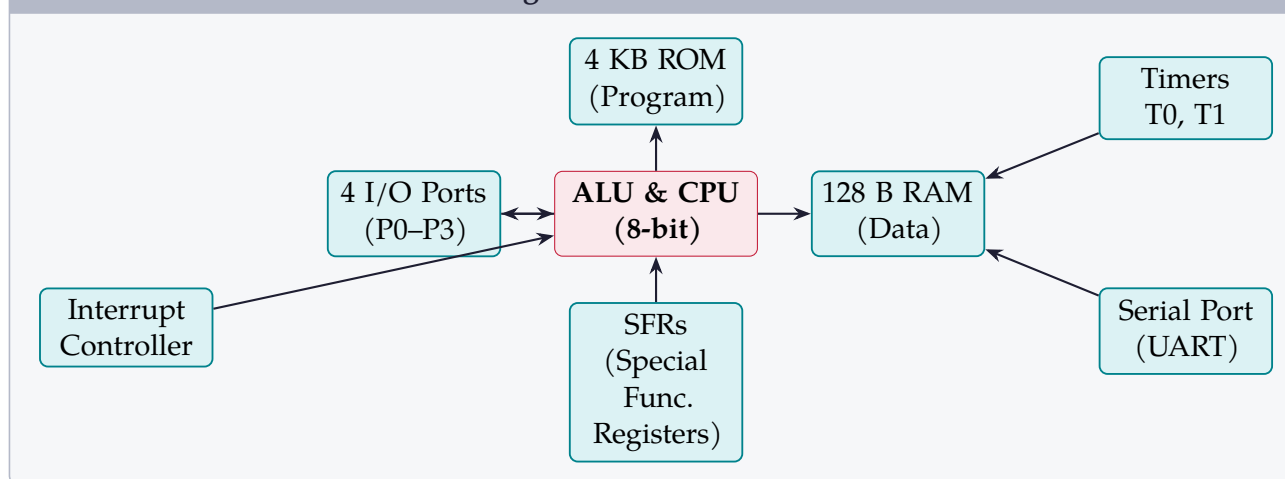
8051 Microcontroller

The **8051** is an 8-bit, single-chip microcontroller with on-chip program memory (ROM), data memory (RAM), I/O ports, timers, a serial port, and interrupt logic. It operates on a Harvard architecture (separate program and data memory buses).

1.1 Key Features of 8051

- **CPU:** 8-bit ALU, accumulator (A), B register, PSW (Program Status Word).
- **ROM (Program Memory):** 4 KB on-chip; expandable to 64 KB externally.
- **RAM (Data Memory):** 128 bytes on-chip; upper 128 bytes are SFRs (Special Function Registers).
- **I/O Ports:** Four 8-bit bidirectional I/O ports (P0, P1, P2, P3).
- **Timers:** Two 16-bit timers/counters (Timer 0 and Timer 1).
- **Serial Port (UART):** Full-duplex serial communication (TXD/RXD on P3).
- **Interrupts:** Five interrupt sources — INT0, INT1, Timer0, Timer1, Serial.
- **Clock:** Typically 12 MHz crystal; one machine cycle = 12 clock cycles.

8051 Internal Architecture Block Diagram



1.2 Real-World Interfacing of 8051

- **LED / Switch:** GPIO pins directly drive LEDs (with current-limiting resistors); switches pulled up to VCC.
- **LCD (16x2):** Driven via P2 (data bus) and P3 control lines (RS, RW, EN).
- **ADC (e.g., ADC0804):** External ADC connected via parallel data bus and control signals.

- **Stepper Motor:** Driven via ULN2003 driver IC connected to P1.
- **Serial Communication:** TXD (P3.1) and RXD (P3.0) connect to MAX232 (RS-232 level converter) for PC communication.

2. ARM Processor Architecture

ARM (Advanced RISC Machine) Processor

ARM is a family of RISC (Reduced Instruction Set Computer) processor architectures developed by ARM Holdings. ARM processors are characterized by a 32-bit (or 64-bit) load-store architecture, fixed-length 32-bit instructions (Thumb: 16-bit compressed), low power consumption, and high code density.

2.1 ARM vs. 8051 — Key Differences

Feature	8051	ARM Cortex-M
Architecture	CISC (8-bit, Harvard)	RISC (32-bit, Harvard)
Registers	A, B, R0–R7 (bank-switched)	16 general-purpose 32-bit registers (R0–R15)
Clock Speed	12 MHz (typical)	Up to 168 MHz+ (Cortex-M4)
Pipeline	Single-stage	3-stage (Cortex-M0) to 6-stage (Cortex-M4)
Instruction Set	Complex, variable-length	RISC + Thumb-2 (16/32-bit)
Built-in DSP	No	Yes (Cortex-M4 has FPU + DSP extensions)
Power	Low (simple core)	Ultra-low to medium
Example Chips	AT89S52, DS89C4x0	STM32, LPC1768, Nordic nRF52

2.2 ARM Processor Features

- **RISC Pipeline:** Multi-stage instruction pipeline (Fetch → Decode → Execute) allows higher throughput.
- **Load-Store Architecture:** Only Load and Store instructions access memory; all other operations work on registers.
- **Thumb Instruction Set:** 16-bit compressed instructions reduce code size by ~30% while maintaining most of the 32-bit performance.
- **Barrel Shifter:** Integrated into ALU — allows free-of-cost shifts/rotates within any data-processing instruction.
- **NVIC (Nested Vectored Interrupt Controller):** Fast, low-latency interrupt handling with configurable priority levels.

- **Cortex-M Variants:** M0 (ultra-low power), M3 (general purpose), M4 (with FPU and DSP), M7 (high performance).

3. Processor and Memory Organization

3.1 Memory Types in Embedded Systems

Memory Type	Characteristics	Use in Embedded
ROM (Mask ROM)	Non-volatile, factory-programmed, read-only	Fixed firmware in mass-produced ICs
Flash	Non-volatile, electrically erasable, page-by-page write	Main program storage in modern MCUs
EEPROM	Non-volatile, byte-wise electrically erasable	Configuration data, calibration parameters
SRAM	Volatile, fast, no refresh needed	CPU data memory, stack, heap
DRAM	Volatile, needs refresh, high density	External memory for complex embedded systems (not MCU)
SDRAM	Synchronous DRAM, higher speed	External RAM for embedded Linux (e.g., Raspberry Pi)

3.2 Harvard vs. Von Neumann Architecture

Feature	Harvard Architecture	Von Neumann Architecture
Memory Buses	Separate instruction and data buses	Single shared bus
Simultaneous Access	Yes — fetch instruction while reading data	No — only one at a time
Speed	Faster (no bus contention)	Slower (bus bottleneck)
Examples	8051, PIC, ARM Cortex-M	Intel x86 PC
Complexity	Higher (dual bus hardware)	Simpler

3.3 Memory Map of a Typical MCU

The memory-mapped addressing scheme of a microcontroller assigns fixed addresses to each memory region:

Typical MCU Memory Map (ARM Cortex-M)

Peripherals (0x40000000–)	0x4000 0000
SRAM (0x20000000–)	0x2000 0000
Flash / Code (0x08000000–)	0x0800 0000
Boot ROM / System Memory	0x1FFF 0000
Vector Table + Option Bytes	0x0000 0000
0x00000000 (Alias region)	

4. Parallelism in Instruction Level

Instruction-Level Parallelism (ILP)

Instruction-Level Parallelism (ILP) is the ability of a processor to execute multiple instructions simultaneously, overlapping their execution stages using techniques such as **pipelining**, **superscalar execution**, and **VLIW** (Very Long Instruction Word).

4.1 Pipelining

Pipelining divides instruction execution into **stages**, so that multiple instructions can be in different stages of execution simultaneously — like an assembly line.

3-Stage Instruction Pipeline (ARM Cortex-M0)

Cycle	1	2	3	4	5
Instr 1	Fetch	Decode	Execute		
Instr 2		Fetch	Decode	Execute	
Instr 3			Fetch	Decode	Execute

4.2 Pipeline Hazards

Hazard Type	Cause	Solution
Data Hazard	Instruction depends on result of a previous instruction still in pipeline	Register forwarding (bypassing), stall/bubble insertion
Control Hazard	Branch instruction changes program flow before pipeline completes	Branch prediction, pipeline flush
Structural Hazard	Two instructions need same hardware resource simultaneously	Resource duplication, stall

5. Processor and Memory Selection

Selecting the right processor and memory for an embedded application requires balancing multiple constraints:

Processor Selection Criteria

- **Processing Speed:** Clock frequency (MHz) and MIPS (Million Instructions Per Second) rating.
- **Word Width:** 8-bit (simple, low-cost) vs. 16-bit vs. 32-bit (high performance).
- **On-chip Peripherals:** Required ADC channels, communication ports, timers.
- **Power Consumption:** Active current (I_{DD}), sleep current ($I_{standby}$) — critical for battery-powered devices.
- **Cost:** Unit price must fit product BOM (Bill of Materials) budget.
- **Toolchain and Ecosystem:** Availability of compilers, debuggers, libraries, and community support.
- **Temperature Range:** Industrial (-40°C to $+85^{\circ}\text{C}$), automotive (-40°C to $+125^{\circ}\text{C}$).

5.1 Memory Selection Criteria

- **Volatile vs. Non-volatile:** Program code needs non-volatile (Flash); variables need volatile (SRAM).
- **Capacity:** Must accommodate program code, constants, variables, stack, and heap with margin.
- **Access Speed:** RAM access time must be compatible with processor clock (no wait states needed).
- **Endurance:** Flash has limited write cycles ($\sim 100,000$ erase/write cycles) — critical for OTA update designs.
- **Interface:** Parallel bus (faster) vs. serial (SPI/I2C Flash — slower but fewer pins).

Quick Revision — Module 2

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Key Formulas at a Glance

Machine Cycle Time (8051)	$t_{mc} = \frac{12}{f_{osc}}$ (12 oscillator periods per machine cycle)
Processing Speed	$MIPS = \frac{f_{clk}}{CPI \times 10^6}$ (CPI = Cycles Per Instruction)
Pipeline Speedup	$S = \frac{t_{no-pipeline}}{t_{pipeline}} \approx N$ (N = number of stages, ideal)
Memory Address Space	Addressable locations = 2^n (n = address bus width in bits)

One-Line Definitions

- **8051:** An 8-bit Harvard-architecture microcontroller with 4 KB ROM, 128 B RAM, 4 I/O ports, and 2 timers.
- **ARM:** Advanced RISC Machine; 32-bit low-power RISC processor widely used in embedded and mobile devices.
- **RISC:** Reduced Instruction Set Computer — few, simple, fixed-length instructions; high clock rate.
- **Harvard Architecture:** Separate buses for program memory and data memory allowing simultaneous access.
- **Von Neumann:** Single shared bus for instructions and data; simpler but slower.
- **Pipeline:** Technique to overlap execution of multiple instructions by dividing into fetch-decode-execute stages.
- **ILP:** Instruction-Level Parallelism — executing multiple instructions in the same clock cycle.
- **Flash Memory:** Non-volatile, electrically erasable memory used to store program code in MCUs.
- **SRAM:** Static RAM — volatile, fast, no refresh; used for data memory and stack in MCUs.
- **SFR:** Special Function Register in 8051 (addresses 80H–FFH) controlling timers, serial port, I/O, interrupts.
- **Thumb Instruction Set:** 16-bit compressed ARM instruction set reducing code size by ~30%.
- **NVIC:** Nested Vectored Interrupt Controller in ARM; handles priority-based, low-latency interrupts.

Frequently Asked / Viva Questions**Q1. What is the architecture of the 8051 microcontroller?**

The 8051 is an 8-bit Harvard-architecture MCU with a 4 KB ROM (program), 128 bytes RAM (data), four 8-bit I/O ports, two 16-bit timers, one UART serial port, and five interrupt sources. One machine cycle = 12 oscillator clock cycles.

Q2. What is the difference between Harvard and Von Neumann architecture?

Harvard uses separate buses for program and data memory, enabling simultaneous fetch and data access (faster). Von Neumann uses a single bus for both (simpler but has bottleneck). 8051 and ARM MCUs use Harvard; x86 PCs use Von Neumann.

Q3. What is pipelining? What is its advantage?

Pipelining divides instruction execution into stages (Fetch, Decode, Execute) and processes multiple instructions concurrently in different stages. Advantage: Increases throughput (instructions completed per clock cycle) without increasing clock frequency.

Q4. Name the three types of pipeline hazards.

- **Data Hazard:** Dependency between instructions (solved by forwarding/stall).
- **Control Hazard:** Branch instructions disrupting pipeline flow (solved by branch prediction).
- **Structural Hazard:** Two instructions compete for same hardware resource.

Q5. Why is ARM preferred over 8051 for modern embedded systems?

ARM (Cortex-M) offers 32-bit data width, higher clock speeds (up to 168 MHz), a pipelined RISC core, Thumb instruction set for code efficiency, built-in NVIC, optional FPU/DSP, and ultra-low power modes. 8051 is only 8-bit and runs at 12 MHz.

Q6. What is the Thumb instruction set in ARM?

Thumb is a 16-bit compressed version of the 32-bit ARM instruction set. It reduces code size by ~30% while retaining most of the performance, making it ideal for memory-constrained embedded systems.

Q7. What are Special Function Registers (SFRs) in 8051?

SFRs are memory-mapped registers (addresses 80H–FFH) that control the MCU's internal peripherals such as ports (P0–P3), timers (TCON, TMOD), serial port (SCON, SBUF), interrupt control (IE, IP), and the stack pointer (SP).

Q8. What is the difference between Flash memory and EEPROM?

Flash memory is erased in large **blocks/pages** and is much faster and higher-density — used for program code storage. EEPROM is erased byte-by-byte (or word-by-word), has fewer erase cycles, but allows fine-grained updates — used for configuration data.

Frequently Asked / Viva Questions (contd.)

Q9. What factors are considered in processor selection for an embedded system?

Processing speed (MIPS), word width (8/16/32-bit), on-chip peripheral set, power consumption, operating temperature range, unit cost, and availability of compilers, development tools, and community support.

Q10. What is memory-mapped I/O?

In memory-mapped I/O, peripheral registers (I/O ports, timers, UART) are assigned addresses in the same address space as RAM. The processor accesses them using regular memory read/write instructions — no separate I/O instructions needed (used in ARM Cortex-M).

Q11. What is the significance of the barrel shifter in ARM?

The barrel shifter in ARM's ALU allows any data-processing instruction to optionally shift or rotate one of its source operands by any amount **in the same clock cycle** as the main operation — effectively providing free shift operations.

Q12. What is MIPS and how is it calculated?

MIPS (Million Instructions Per Second) measures processor performance. $MIPS = f_{clk} / (CPI \times 10^6)$ where CPI is Cycles Per Instruction. For 8051 at 12 MHz with CPI = 12: $MIPS = 12 / (12) = 1$ MIPS.

8051 vs. ARM Cortex-M — At a Glance

Parameter	8051	ARM Cortex-M4
Data Width	8-bit	32-bit
Max Clock	12 MHz (standard)	168 MHz
Architecture	Harvard, CISC-like	Harvard, RISC
Pipeline	None	6-stage
On-chip Flash	4 KB (original)	Up to 2 MB
On-chip SRAM	128 bytes	Up to 256 KB
FPU	No	Yes (single precision)
Interrupts	5 sources	Up to 240 (NVIC)